TOPICS IN ERROR-CORRECTING CODES

Practice midterm

Date: 9 November 2026 **Time:** 75 minutes

Name: _____ **MIT ID:** _____

10 points per problem; final score = min(total points, 60).

Problem 1 **10 points**

Rollback in interactive coding. In the computationally efficient interactive coding scheme from class, a decoded header reports that the other party's transcript is much shorter. Why does the receiving party delete at most one block in the round?

Solution.

Problem 2 **10 points**

Distinct labels in a tree code. Suppose the two edges leaving each internal vertex of a binary tree have distinct labels. Does this guarantee a positive lower bound on relative tree-code distance that is independent of the depth? Justify your answer.

Solution.

Problem 3 **10 points**

Systematic Reed–Solomon encoding. Given distinct evaluation points $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ and $1 \le k \le n$, design an explicit map f from $\mathbb{F}_q^k$ to polynomials of degree at most $k-1$ that results in a systematic Reed–Solomon code. In particular, for every message $\mathbf{m} \in \mathbb{F}_q^k$, if the corresponding polynomial is $f_{\mathbf{m}}(X)$, then the vector $\left(f_{\mathbf{m}}(\alpha_i)\right)_{i \in [n]}$ must have the message $\mathbf{m}$ in its first k positions.

Solution.

Problem 4 **10 points**

Generalized Singleton bound for $L = 3$. Let $C_n \subseteq \mathbb{F}_q^n$ be a family of codes whose rates tend to $R > 0$. Suppose every Hamming ball of relative radius ρ contains fewer than $L = 3$ codewords. Prove that

$$\rho \leq \frac{2}{3}(1 - R).$$

Solution.

Problem 5 **10 points**

Bounding right degrees in an expander. Let $G = (L, R, E)$ be a bipartite graph with $|L| = n \geq 1$ and $|R| = m$. Each left vertex has degree $c \geq 1$. Suppose $|\Gamma_G(S)| \geq \gamma|S|$ for all $S \subseteq L$ with $|S| \leq \alpha n$. Construct a bipartite graph $G' = (L, R', E')$ with the same left degree and expansion guarantee, such that

$$m \leq |R'| \leq 2m \qquad \text{and} \qquad \deg_{G'}(r') \leq \left\lceil \frac{nc}{m} \right\rceil \quad \text{for every } r' \in R'.$$

Solution.

Problem 6 **10 points**

Distance amplification by tuples. Let $C \leq \mathbb{F}_q^n$ be a linear code of rate R and relative minimum distance δ. For every integer $m \geq 1$, construct a code over $\mathbb{F}_q^m$ with block length n^m, rate

$$\frac{R}{mn^{m-1}},$$

and relative minimum distance $1 - (1 - \delta)^m$.

Solution.

Problem 7 — 10 points

Neighborhoods and coordinate subspaces. Let $G = ([m]_L, [m]_R, E)$ be bipartite. Write $N(j)$ for the left neighbors of a right vertex j, and $N(I)$ for the right neighbors of $I \subseteq [m]_L$. Over a field $\mathbb{F}$, define

$$U_I = \operatorname{span}\{e_i : i \in I\}, \qquad H_j = \left\{ x \in \mathbb{F}^m : \sum_{i \in N(j)} x_i = 0 \right\}.$$

(a) Prove that for every nonempty $I \subseteq [m]_L$,

$$\sum_{j=1}^{m} \big(\dim U_I - \dim(U_I \cap H_j)\big) = |N(I)|.$$

(b) Construct a graph $G = ([m]_L, [m]_R, E)$ together with a nonzero subspace $W \leq \mathbb{F}^m$, such that G has a perfect matching and $W \subseteq H_j$ for every right vertex j. Specify m, the edges, a perfect matching, and a basis of W.

Solution.

Problem 8 **10 points**

Formal derivatives and multiplicity codes. Let $d \geq 1$ and let $\mathbb{F}$ be a field with $\mathrm{char}(\mathbb{F}) > d$. Write $P^{(i)}$ for the ith formal derivative of P.

(a) For $P \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$, prove

$$P^{(i)}(\alpha) = 0 \ (0 \leq i < d) \quad \iff \quad (X - \alpha)^d \mid P.$$

Hint: Apply the chain rule to $P(X + \alpha)$.

(b) Let $\mathbb{F}$ be finite, let $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ be distinct, and let $1 \leq k \leq dn$. Prove that the order-d code $\mathrm{UMULT}_{\mathbb{F}}((\alpha_1, \ldots, \alpha_n), k)$ has relative minimum distance at least

$$1 - \frac{1}{n} \left\lfloor \frac{k-1}{d} \right\rfloor.$$

Solution.